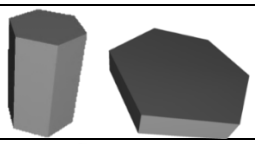
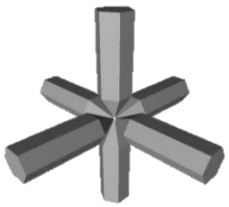
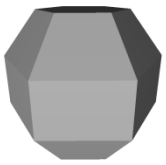
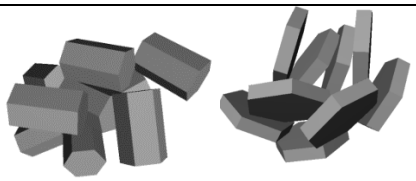
Default types of particles

In table below there is description of the default particles and their parameters.

All parameters except one are floating point data type. Exception is the “number of particles” for aggregate particle type, it has integer data type.

Abbreviations for some parameters:

1. Length (*L*)
2. Long diameter of the base of the particle (*D*)
3. Peak height (*h*)

Value	Particle name	3D-model	Parameter list
1	Hexagonal prism		1. *L* 2. *D*
2	Bullet		1. *L* 2. *D* 3. *h*
3	Bullet-rosette		1. *L* 2. *D* 3. *h*
4	Droxtal		1. Top angle 2. Bottom angle 3. Radius
5-9	*disabled*	-	-
10	Hollow-column		1. *L* 2. *D* 3. Cavity angle
11	*disabled*	-	-
12	Aggregate of the hexagonal prism		1. *L* 2. *D* 3. Number of particles

Input parameters

Notes:

1. All angles should be in degrees.
2. Size parameters of particle and wavelength should be the same unit (micrometers or millimeters or nanometers etc.).

Main parameters (mandatory):

Description of parameter	Command line key	No.	Description of values	Data type	Allowed values
Particle (can be put with 2 ways)	**-p 1 2...**	**1**	number of the default particle	Integer	1, 2, 3, 4, 10, 12
		2	list of the default particle parameters (see previous table)	-	-
	--pf -p 1	**1**	file name of the particle	String	-
Refraction index	**--ri 1 2**	**1**	real part (Re)	Floating point	-
		2	imaginary part (Im) (if no Im part then should be "0")	Floating point	-
Max. number of internal refractions	**-n 1**	**1**	number of internal refractions	Integer	0-20
Number of spatial orientation of particles (can be put with 2 ways)	**--random 1 2**	**1**	number of steps by zenith angle (β)	Integer	Positive number
		2	number of steps by azimuth angle (γ)	Integer	Positive number
	--fixed 1 2	**1**	angle β	Floating point	-
		2	angle γ	Floating point	-
Method flag (can be put with 2 ways)	**--po**	**-**	physical optics approximation	-	-
	--go	**-**	geometrical optics approximation	-	-
Grid of the direction scattering sphere (DSS) (can be put with 2 ways)	**--grid 1 2 3**	**1**	max. scattering zenith angle (θ in DSS coordinates), it counts from the backscattering direction (180°), i.e. if it is "180" then scattering angle range will be [0°-180°].	Floating point	0-180
		2	number of steps by azimuth angle (φ in DSS coordinates)	Integer	Positive number
		3	number of steps by θ	Integer	Positive number
	--grid 1 2 3 4	**1**	start θ angle of scattering range	Floating point	Positive number
		2	end θ angle of scattering range	Floating point	Positive number
		3	number of steps by φ	Integer	Positive number
		4	number of steps by θ	Integer	Positive number
Wavelength	**-w 1**	**1**	wavelength	Floating point	Positive number
All angles flag (need for phys. optics)	**--all**	**-**	-	-	-

Additional parameters (optional):

Description of parameter	Command line key	No.	Description of values	Data type	Default value
Output file name	**-o 1**	**1**	file name (without extension and parentheses)	String	"M"
Spatial orientation range of particle by β	**-b 1 2**	**1**	start β angle	Floating point	0
		2	end β angle	Floating point	180
Spatial orientation range of particle by γ	**-g 1 2**	**1**	start γ angle	Floating point	0
		2	end γ angle	Floating point	360
Resize of particle parameter	**--rs 1**	**1**	new size (D_{max}) of particle (in case of aggregate it is new size of single particles)	Floating point	-
Scattered beam filter	**--filter 1**	**1**	max. scattering angle θ, it counts from the backscattering direction (180°), beams with direction out of [angle-180°] range will be ignored.	Floating point	-
Error orientation filter	**-r**	**1**	ratio for filtering orientations with errors of beam overlays	Floating point	100

Flags that are disabled by default (optional):

Description of parameter	Command line key
Enable absorption	**--abs**
Program should be closed after the calculation	**--close**
Disable coherence	**--incoh**
Disable "shadow beam"	**--shadow_off**
Change to procedure for convex particle forcefully	**--forced_convex**
Change to procedure for non-convex particle forcefully	**--forced_nonconvex**

Examples of commands:

1. Hexagonal column, geometrical optics, all scattering angles:
   -p 1 100 69.6 --ri 1.3116 0 –n 8 --random 300 400 --go --grid 180 45 360 --all –w 0.532 –o out
2. Particle from file "cube1.dat", physical optics, backscattering direction, absorption enabled, the program should be closed, with scattering filter of 40°:
   --pf -p cube1.dat --rs 10 --ri 1.32 0.000019 -n 12 --random 1200 750 --po --grid 0.5 2 2 --all –w 1.064 –o out2 --abs --close --filter 40
3. To do multithread calculations you just need to split your calculation by orientations of beta or gamma angle. In that case you need to use **"-b"** or **"-g"** parameter. For example:
   -p 1 100 69.6 --ri 1.3116 0 –n 8 --random 150 400 --po --grid 0.5 2 2 --all –w 1.064 –o out2 --abs --close -b 0 45
   -p 1 100 69.6 --ri 1.3116 0 –n 8 --random 150 400 --po --grid 0.5 2 2 --all –w 1.064 –o out2 --abs --close -b 45.3 90

Algorithm in short

1. Choose method (Physical Optics approximation / Geometrical optics appr.)
2. Set particle (Default shape / From file)
3. Trace
    a. Split beams (Convex / Non-convex)
    b. Scatter beams
        i. Diffraction / No diffraction
        ii. Calculate matrices (Mueller's matrices)
4. Output matrices (in file).